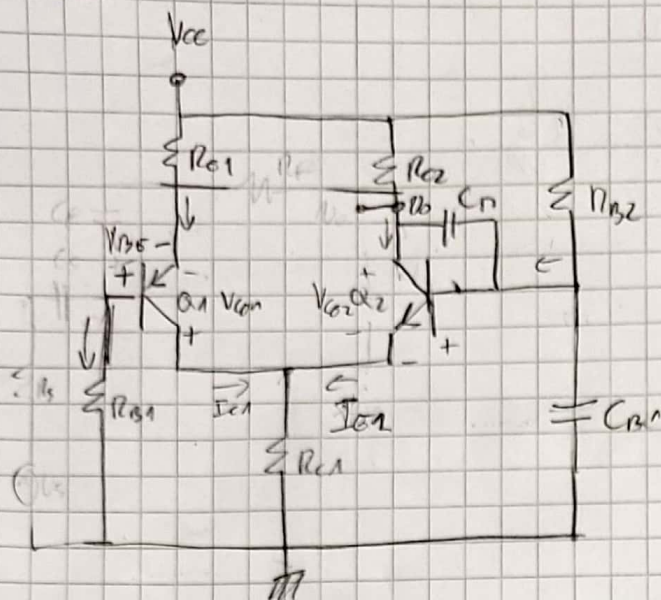


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1 op point

$$V_C - R_{C1} I_{C1} + V_{BE1} - R_{E1} I_{B1} = 0$$

$$V_{CC} - R_{C1} (1 + \beta_{F1}) I_{B1} + V_{BE1} - R_{E1} I_{B1} = 0$$

$$I_{B1} = \frac{V_{CC} + V_{BE1}}{R_{C1} + R_{E1} (1 + \beta_{F1})} = 6,97 \mu A \quad I_{C1} = \beta_{F1} I_{B1} = \boxed{0,745 \text{ mA}} \quad I_{E1} = 0,75 \text{ mA}$$

$$V_{CC} - R_{C2} I_{C2} - V_{BE2} - R_{E1} (I_{C1} + I_{E2}) = 0$$

$$V_C - R_{C2} I_{C2} - V_{BE2} - R_{E1} I_{C1} - R_{E1} (1 + \beta_{F2}) I_{B2} = 0$$

$$I_{B2} = \frac{V_C - V_{BE2} - R_{E1} I_{C1}}{R_{C2} + R_{E1} (1 + \beta_{F2})} = 5,03 \mu A \quad I_{C2} = \beta_{F2} I_{B2} = \boxed{1,006 \text{ mA}} \quad I_{E2} = 1,01 \text{ mA}$$

$$V_{CC} - R_{E1} I_{E1} + V_{CE1} - R_{E1} (I_{C1} + I_{E2}) = 0$$

$$V_{CE1} = -V_{CC} + R_{E1} I_{E1} + R_{E1} (I_{C1} + I_{E2}) = \boxed{-7,89 \text{ V}}$$

$$V_{CC} - R_{C2} I_{C2} - V_{CE2} - R_{E1} (I_{C1} + I_{E2}) = 0$$

$$V_{CE2} = V_{CC} - R_{C2} I_{C2} - R_{E1} (I_{C1} + I_{E2}) = \boxed{3,51 \text{ V}}$$

$$g_{m1} = 28,65 \text{ mS}^{-1}$$

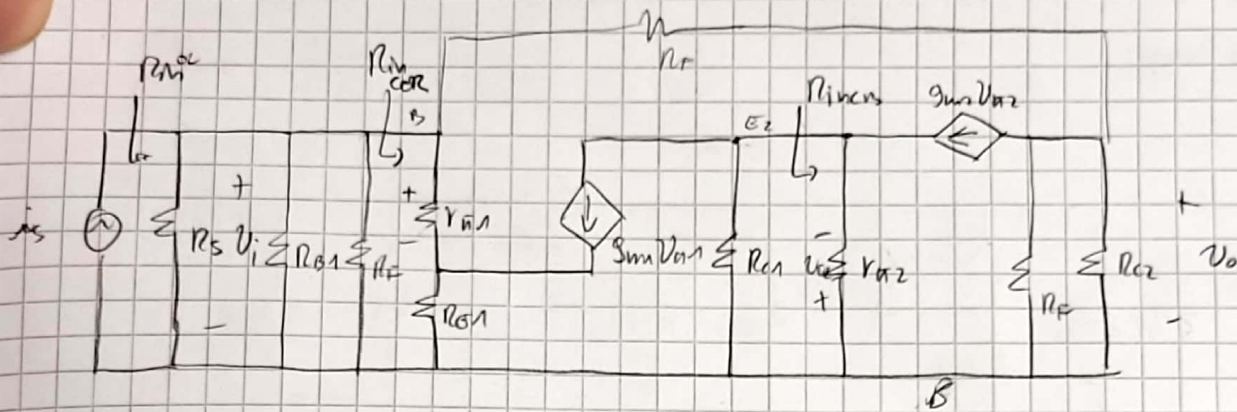
$$g_{m2} = 38,17 \text{ mS}^{-1}$$

$$r_{\pi 1} = 5,23 \text{ k}\Omega$$

$$r_{\pi 2} = 5,18 \text{ k}\Omega$$

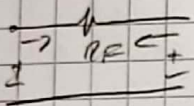
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2) $A_v = V_o / V_s$ We have no voltage sampling current mixing $\rightarrow$ voltage at source



Protonen $\begin{cases} i_E = \beta I_B + V_F / R_{E1} \\ i_O = V_o / R_o \end{cases}$

$\beta = -\frac{1}{R_F} = -4,545 \cdot 10^{-5} \Omega^{-1}$ $R_{E1} = R_o = R_F = 22 \text{ k}\Omega$



$R_{in2} = \frac{r_{\pi 2}}{1 + \beta_{o2}} = 25,77 \Omega$ $R_{in2} = 76,2 \text{ k}\Omega$

$A_v^{OL} = A_{vCB} \cdot A_{vCE} = \frac{V_o}{V_i}$

$A_{vCB} = \frac{V_o}{V_{i2}} = g_{m2} (R_F \parallel R_{C2}) = 149,87$

$A_{vCE} = \frac{V_{i2}}{V_i} = \frac{\beta_{o1} (R_{C1} \parallel R_{in2})}{r_{\pi 1} + R_{E1} (1 + \beta_{o1})} = -0,0495$

$\frac{V_i}{V_s} = R_s \parallel R_{B1} \parallel R_{B2} \parallel R_{in1} = 0,944 \text{ k}\Omega$

$A_{vol} = -7 \text{ k}\Omega$ $A_v^F = \frac{A_v^{OL}}{1 + \beta_{o1}} = 5,3 \text{ k}\Omega$
 $|1 + \beta_{o1}| = 531 \text{ k}\Omega$

voltage gain

$\frac{V_s}{V_o} = \frac{1}{R_s}$ $A_v = A_v^F \cdot \frac{1}{R_s} = 5,3$

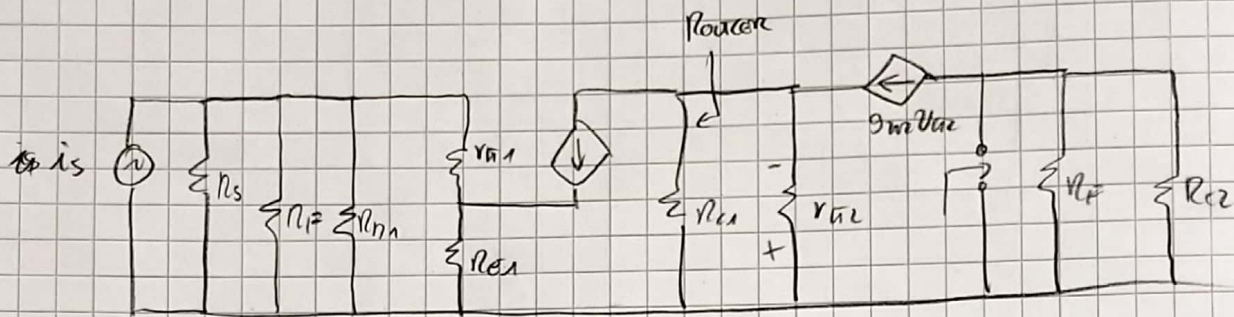
$R_{in}^{OL} = \frac{V_i}{I_s} = 0,944 \text{ k}\Omega$ $R_{in}^F = R_{in}^{OL} / (1 + \beta_{o1}) = 0,716 \text{ k}\Omega$ $R_{out}^F = \frac{R_F \parallel R_{C2}}{1 + \beta_{o2}} = 2,936 \text{ k}\Omega$

3) $R_{in} = \frac{1}{\frac{1}{R_{in}^F} - \frac{1}{R_s}} + R_s = 1,077 \text{ k}\Omega$

4) $R_{out} = R_{out}^F = 2,936 \text{ k}\Omega$

5) Upper limit considers only C_n

R_{out} (so R_i^o) is affected by feedback (is reduced) so we can use feedback theory

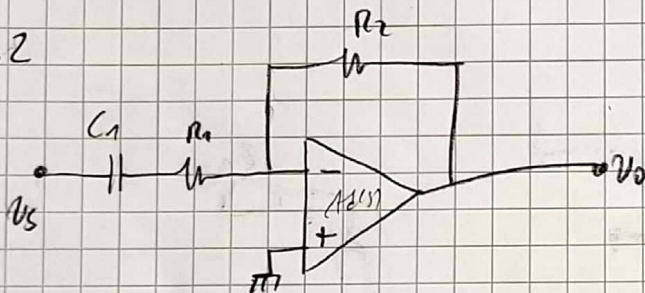


$$R_{out}^o = R_L // R_{out} = 3.87 \text{ k}\Omega$$

$$f_H^u = \frac{1}{2\pi} \frac{1}{C_n R_{in}^o} = 4.11 \text{ MHz}$$

$$f_H^F = f_H^u (1 + T) = 5.42 \text{ MHz}$$

Ex 2



$$R_2 = 22 \text{ k}\Omega$$

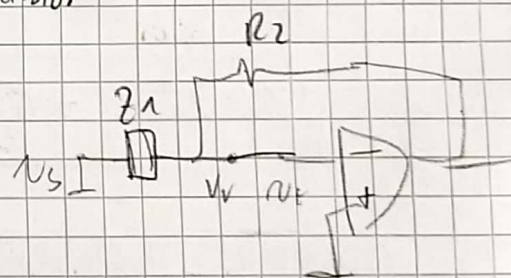
$$C_1 = 100 \text{ nF}$$

$$A_d(s) = \frac{A_{d0}}{1 + s/\omega_0}$$

1) Considering $R_1 = 0 \Omega$ calculate $A_v(s) = v_o/v_s(s)$

$$A_v(s) = \frac{Z_2(s)}{Z_1(s)} = \frac{-R_2}{\frac{1}{sC_1}} = -sC_1 R_2 \quad \text{ideal differentiator}$$

2) estimation of phase margin



$$T = \frac{v_o}{v_v} = A_d(s) \cdot \frac{1/sC_1}{R_2 + 1/sC_1} = \frac{\frac{A_d(s)}{sC_1}}{\frac{1 + sC_1 R_2}{sC_1}} = \frac{A_{d0}}{1 + s/\omega_0} \cdot \frac{1}{1 + sC_1 R_2}$$

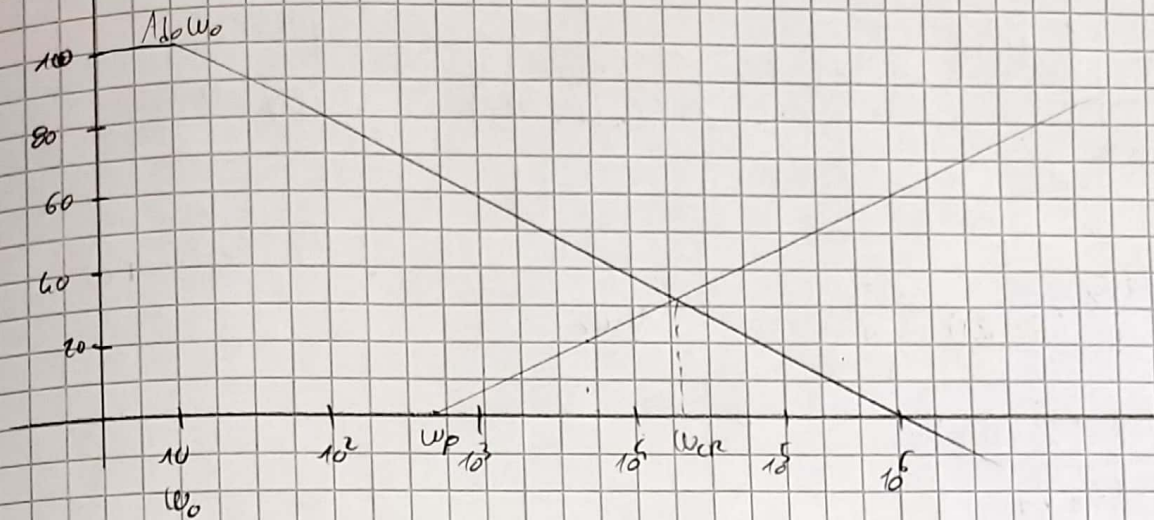
$$\beta(s) = \frac{1}{1 + sC_1 R_2}$$

$$\beta(0) = 1 = 0 \text{ dB}$$

$$\beta(\infty) = 0$$

$$\frac{1}{\beta(s)} = 1 + sC_1 R_2$$

$$\omega_p = \frac{1}{C_1 R_2} = 454 \text{ rad/s}$$



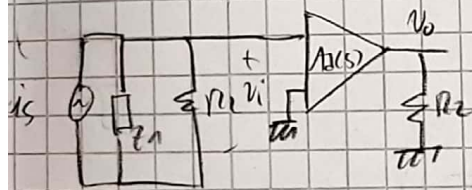
$$\begin{cases} A_{db} \omega_0 = A_x \omega_{cr} \\ \frac{1}{\omega_p} = \frac{A_x}{\omega_{cr}} \end{cases} \Rightarrow \begin{cases} \frac{\omega_{cr}^2}{\omega_p} = A_{db} \omega_0 \\ A_x = \frac{\omega_{cr}}{\omega_p} \end{cases} \quad \omega_{cr} = \sqrt{A_{db} \omega_0 \omega_p} = \sqrt{2,13 \cdot 10^9} \text{ rad/s}$$

$$f_{cr} = \frac{\omega_{cr}}{2\pi} = 3,4 \text{ kHz}$$

$$\varphi_m = 180 - 2 \arctan\left(\frac{\omega_{cr}}{\omega_0}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_p}\right) = 112,5^\circ$$

3) block diagram of sys. according to feedback theory

$$R_1 R_2 = R_2 \quad \beta = -\frac{1}{R_2}$$



$$A_v = \frac{v_o}{v_s} = \frac{v_o}{v_i} \cdot \frac{v_i}{v_s} = A_d(s) \cdot \frac{v_i}{v_s} = A_d(s) \cdot \left(\frac{R_1}{R_1 + R_2} \right)$$

$$= A_d(s) \left(\frac{1}{1 + \frac{R_2}{R_1}} \right)$$

$$= A_d(s) \frac{R_1}{1 + \frac{R_2}{R_1}}$$



$$1) \quad \beta = -\frac{1}{R_2}$$

2) value of R_2 to make

$$\varphi_m = 60^\circ$$

5) Value of R_1 to move $\angle m = 60^\circ$

introducing R_1 : introduces a zero and move slightly the pole

$$T = A_d(s) \cdot \frac{\frac{1}{sC_1} + R_1}{R_2 + R_1 + \frac{1}{sC_1}} = A_d(s) \cdot \frac{1 + sC_1 R_1}{1 + sC_1 (R_2 + R_1)}$$

$$\phi_m = 60^\circ$$

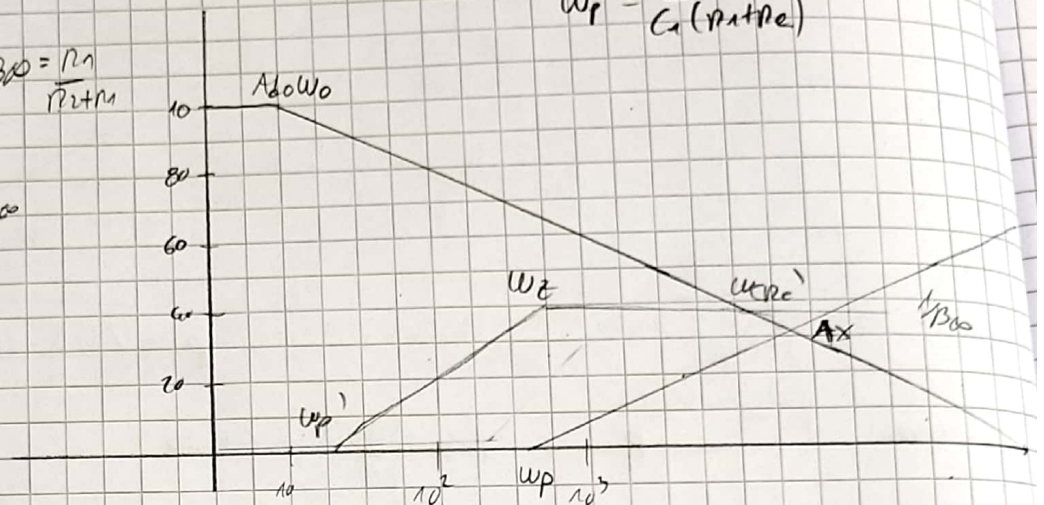
$$\omega_z = \frac{\omega_{enc}}{4}$$

$$\omega_p' = \frac{1}{C_1 (R_2 + R_1)}$$

$$B_o(s) = \frac{1 + sC_1 R_1}{1 + sC_1 (R_2 + R_1)} \quad B_o = \frac{R_1}{R_2 + R_1}$$

$$\frac{1}{B_o}(s) = \frac{1 + sC_1 (R_2 + R_1)}{1 + sC_1 R_1} \infty$$

We now look for ω_{enc}
(after comp)



$$\begin{cases} A_{d0} W_0 = A_x \omega_{enc} \\ \frac{1}{\omega_p'} = \frac{A_x}{\omega_z} \end{cases} \Rightarrow \begin{cases} A_{d0} W_0 = \frac{\omega_z \omega_{enc}}{\omega_p'} \\ A_x = \frac{\omega_z}{\omega_p'} \end{cases} \Rightarrow A_{d0} W_0 = \frac{\omega_{enc}^2}{\omega_p'} \quad \omega_p' = \omega_p = 456 \text{ rad/s}$$

$$\omega_{enc} = \sqrt{A_{d0} W_0 \omega_p'} \approx \omega_{enc} = 218 \cdot 10^4$$

$$\omega_z = \frac{\omega_{enc}}{4} = 5450 \text{ rad/s}$$

$$R_1 = \frac{1}{\omega_z C_1} = \boxed{1739 \Omega}$$

$$\angle m = 180 - \arctan\left(\frac{\omega_{enc}}{\omega_0}\right) - \arctan\left(\frac{\omega_{enc}}{\omega_p'}\right) + \arctan\left(\frac{\omega_{enc}}{\omega_z}\right) = \boxed{76,14^\circ}$$